

Lightweight Soaring

by Brian Creer
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Since he gave up flying Liberators and took up gliding, the main ambition of Keith Jarvis has been to own his own aircraft. Keith is small and built his aircraft - Joey - the same way 32 ft span on a 12 ft fuselage.

Into this shape I was recently installed to do some development flying on the machine which had previously done only hops and short circuits.

Joey, I found, was somewhat larger inside than had appeared although it was necessary for me to remove my shoes in order to get rudder movement.

The cockpit is laid out on simple but functional lines.

Lef to right the instrument panel runs : takeoff dolly release knob, cable release knob, A.S.I., rate of climb beneath which is a bubble-type slip indicator and an altimeter.

The wooden frame canopy fits closely around the pilot's head and into the centre section gap cover and as yet has no ventilation.

Having squeezed me into the cockpit and confessed he could,'t give me any advice on how to fly it, Keith let me take his pride and joy away with the help of Cliff Gurr's Austin Healy on the other end of the string.

The Healy/Joey power/weight ratio was so high that by the time I had eased off the deck, dropped the wheels and carefully felt into a climb, the A.S.I. was reading 60 mph and as it continued to increase I released at 600 feet.

During the initial stage of the take-off and climb, I was impressed by the rapidity of the control actions and of the solid feel of the machine as a whole.

This general impression persisted throughout the rest of the flight -- your were not in a small aircraft but rather in a normal aircraft with a small cockpit -- there is no trace of lightness in the way the aircraft flies.

I had no sooner dropped the wire and assumed level flight, which is rather nose-down from the pilot's point of view, than the rate of climb jerked up and I began circling in quite good lift and Joey started going up -- as did my estimation of the beautifully harmonised controls.

I once wrote that the ailerons on the Kookaburra are the way most pilots dreamed about, but since flying Joey I can say that this machine has the Kookaburra licked pants down in this respect.

The thermal was quite easy to work and gave me ample time to settle down with the strange machine and in settling down I found that there was nothing strange about it, it was an ordinary sailplane except, as I have said, with a small cockpit.

My stockinged toes could easily curl over the little rudder pedals. At 3800 feet I left the lift -- or lost it -- and decided to see what the little machine would do.

First I tried rolling from one side to the other while flying straight and level and found that the aircraft would roll as fast as the pilot could move the stick, and there was little tendency to adverse yaw.

Steep turns were next in the book and some idea of the rapid response to a quick aileron movement can be gained from the fact that on my first attempt at a fast reversal turn, I half-rolled into an inverted position.

Then came the next big surprise, for in recovering from this bad fumble the ASI wound itself right off the click (which ends at 65 mph) and continued on it's second round up to the 30 mark. I can only assume therefore, that the penetration is good!

Joey stalls at an indicated 30 mph and the stall is clean with no tendency for either wing to drag.

In a spin the wing drops sharply, the aircraft rolls, down goes the nose and auto-rotation begins.

Almost immediately speed builds up and the machine becomes un-stalled and tends to flatten into a spiral dive.

I tried to hold the spin by crossing the controls fully once an entry had been made but the same automatic recovery took place. Indeed, the only difficulty in recovering from a spin appears to keep the speed down.

I wasted the better part of the remaining 1800 feet by trying to induce a spin from variously fumble turns and from a side-slip but all with no success.

Vision over the low nose is very good, but the rearward vision is obviously restricted for pilots any larger than Joeys' builder.

The rate of sink was hard to compute on the day this flight was made but it seems to be less than that of the Grunau while the best gliding angle seems to be close to the figure of 1:23 computed for this aircraft.

These impressions are very sketchy since they were obtained from only one flight of about 45 minutes.

This was the first time the machine had been in the air for more than a few minutes at a time, but I am sure that subsequent flights will show that in this small sailplane Keith has found a completely practical aircraft which is light (140 lbs empty weight), not too complicated to build and easily rigged and de-rigged while still having a performance capable of worrying many high-performance pilots.

Anyway, if nothing else, this machine with it's fantastic control would be more than a match for the small, fast summer thermals.

From the Flight Log Book of Keith Jarvis

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Brian Creer

Spins, stalls, thermal flight 4000', auto tow to 600'

45 minutes